

Abutting Tokyo: AI-Driven Cognitive Distortion in VR Urban Visualization

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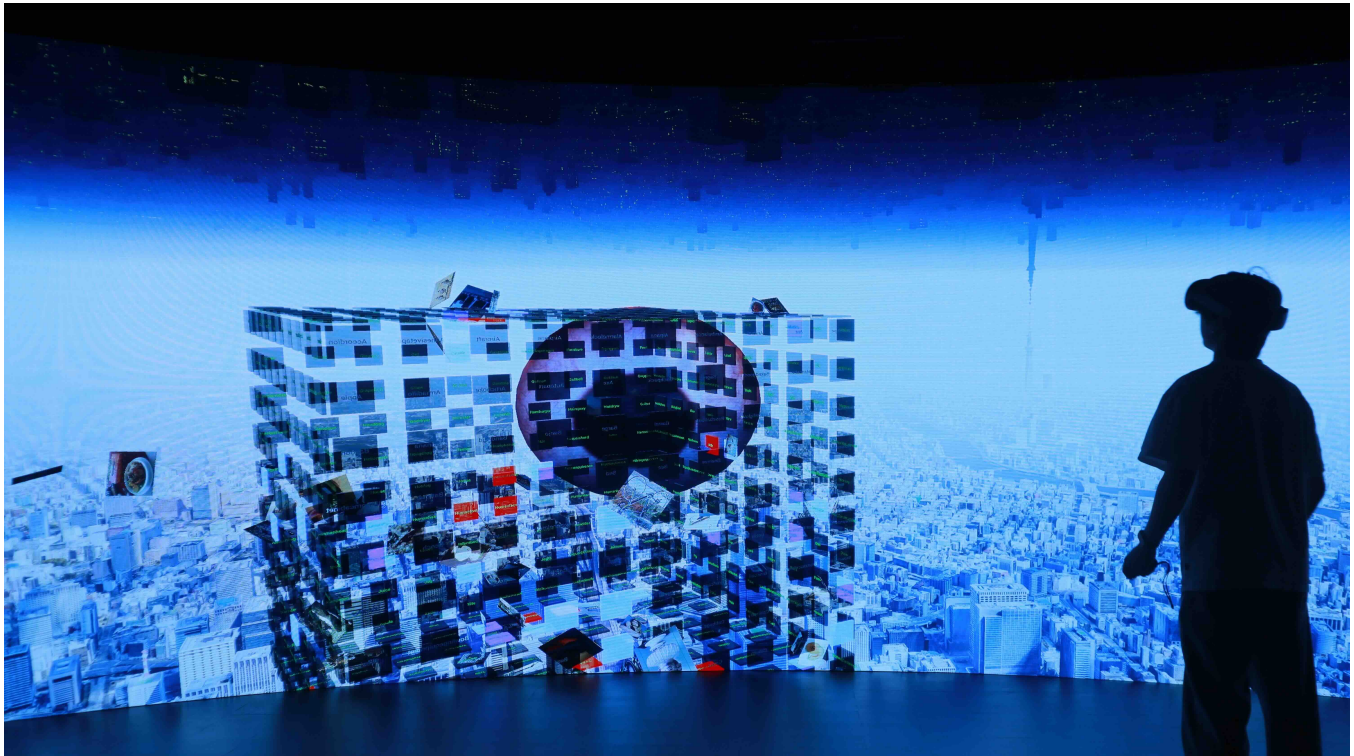


Figure 1: A participant engages in the experience using VR equipment.

Abstract

This paper explores the intersection of artificial intelligence and virtual reality in visualizing urban data. It examines how AI processes and distorts urban data, influenced by the inherent biases in human-annotated datasets. "Abutting Tokyo" presents a VR experience that simulates data annotation process and AI's cognitive distortions, demonstrating the symbiotic relationship between AI cognition and human perception. By critically analyzing AI's interpretation of urban landscapes, the study underscores the impact of data biases and manipulation on AI and human cognition, offering new insights into the dynamics of AI-driven urban visualization.

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CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → *Information visualization*; • **Computing methodologies** → *Philosophical/theoretical foundations of artificial intelligence*; Modeling and simulation.

Keywords

Artificial Intelligence, Virtual Reality, Urban data visualization, Simulation, Cognition

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1 INTRODUCTION

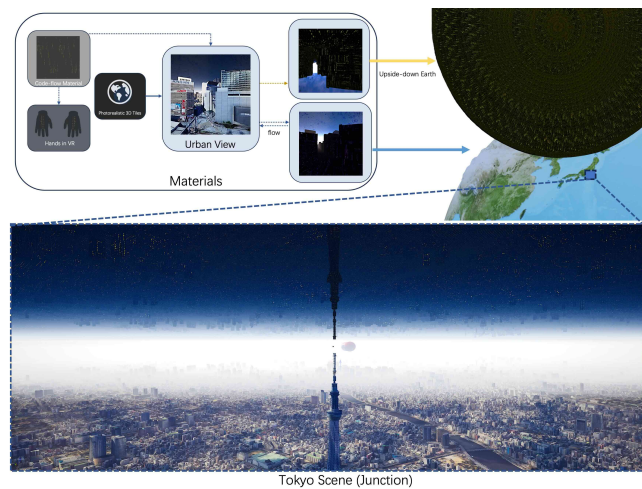


Figure 2: Tokyo's landscape and material realization at the junction of two earths.

The convergence of artificial intelligence and virtual reality is reshaping our understanding of urban environments. The surge in data has altered how we perceive and interact with our surroundings. Artists and technologists have long explored this intersection, creating immersive experiences that blend digital data with physical spaces. These experiences utilize urban data and AI to create dynamic visualizations, challenging our perception of reality.

This paper examines the interactive visualization of Tokyo's urban data through AI, delving into the cognitive boundaries of AI, which are shaped primarily by the datasets used in its training. As scholars such as Kate Crawford have emphasized [Crawford 2021], biases in these datasets can exacerbate social inequalities. This study expands on these discussions by investigating how AI perceives and distorts urban data through VR simulations.

"Abutting Tokyo" presents an innovative VR experience that simulates the data annotation process, using street-view data and a virtual urban environment to explore the cognitive distortions of AI. The term "abutting" refers to the proximity of AI's cognitive

world to the human cognitive world in the data space, much like an upside-down Tokyo that is symmetrical yet not entirely consistent, as shown in Figure 2. Through practice-based research, this paper critically examines the symbiotic relationship between AI cognition, urban data, and human cognition.

2 BACKGROUND AND MOTIVATION

2.1 Data and Space

Data now saturates every facet of our world, leading to an ever-increasing blurring of the lines between physical and virtual spaces. Ascott [Ascott 2003] pioneered the description of a virtual space occupied, created, and sustained by data. Through interactive installations, virtual reality, and augmented reality technologies, audiences can experience unprecedented levels of immersion and engagement in a dynamic environment driven by data.

Artists have long been exploring the integration of physical and digital spaces and the transmission of data. One of the earliest examples is Jeffrey Shaw's *Legible City* [Shaw 1989] which explores the relationship between textual data and physical space. Stanza's *The Emergent City* [Stanza 2007] collects real-time data through urban sensors and transforms this data into dynamic visuals, in order shows the real-time data flow of the city. Refik Anadol's *Machine Hallucinations* [Anadol 2019] leverages the combination of urban data and artificial intelligence technologies. In this work, images data from the city is translated into generated images information by AI, further blurring the boundaries between reality and the virtual.

2.2 AI Cognition

The cognitive boundaries of AI are defined by datasets. The rapid development of machine learning heavily relies on vast amounts of human-annotated data. Crawford argues that these datasets largely determine AI's cognitive limits [Crawford 2021]. However, human annotations inevitably introduce subjective biases and discrimination. Crawford and Paglen's project *Excavating AI* [Crawford and Paglen 2019] exposes these biases in datasets, which can exacerbate societal injustices in practical applications.

Steyerl contends that image and video datasets both reflect and shape social realities [Steyerl 2012]. Connecting data annotation to Saussure's semiotics offers deeper insights into this phenomenon [de Saussure 1916]. The limitations of linguistic symbols in expressing meaning become evident in machine learning processes. AI models, learning from annotated data, are affected by the quality and diversity of training data. Human annotators' subjective biases are amplified during model training, influencing AI's decisions. This relates to Searle's Chinese Room argument [Searle 1980], questioning whether machines truly understand language or merely simulate understanding.

AI artists explore the relationship between AI and human cognition through dataset and algorithm analysis. Paglen's *From 'Apple' to 'Anomaly'* [Paglen 2019] examines potential biases in image recognition using the ImageNet dataset. Vijgen's *The Future Through Artificial Eyes* [Vijgen 2020] investigates AI's impact on our perception and anticipation of the future.

In "Abutting Tokyo," we extend the premise discussed in previous theories and works that consider data as an extension of

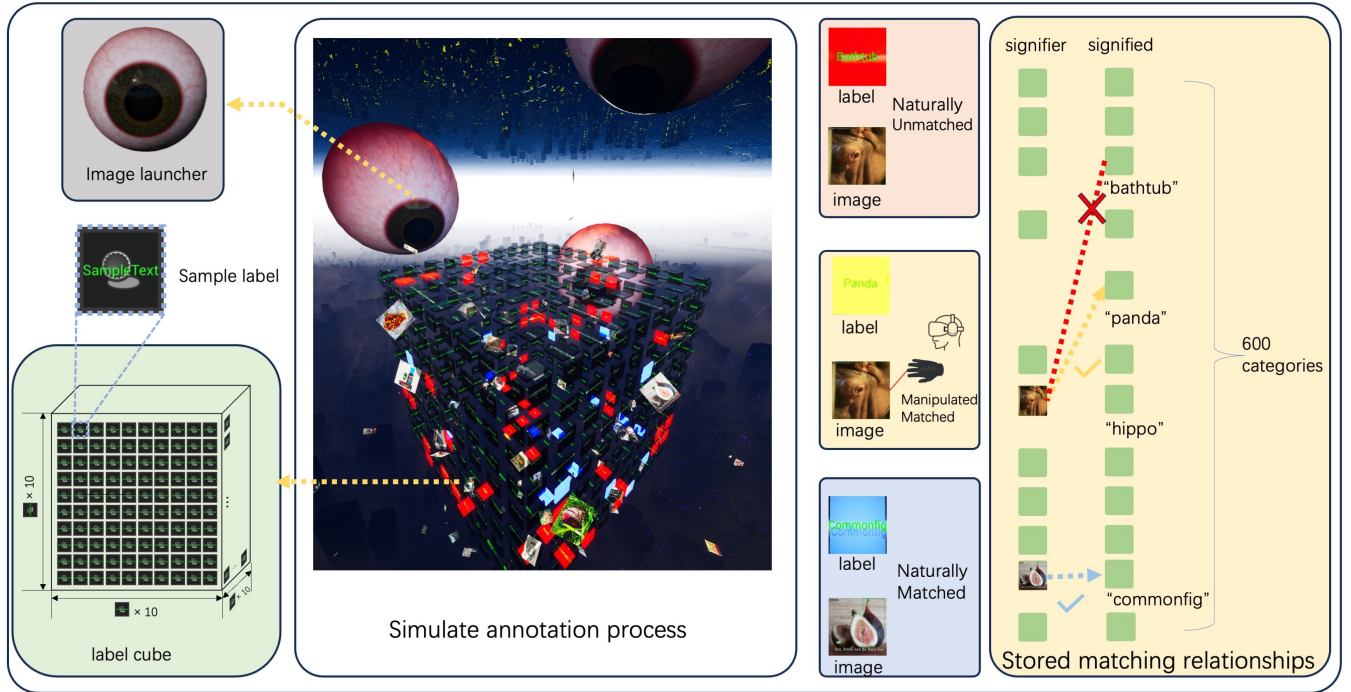


Figure 3: Simulated annotation process. The central cube is composed of stationary 600 labels, corresponding to the 600 categories in dataset. Three giant eyeballs act as image launchers. On the right side are examples of matched and unmatched samples.

human cognition. We aim to critically discuss the interconnections between AI cognition, urban environments, and human cognition through practice-based research, combining data spaces and artificial intelligence as the main thread.

3 SIMULATION AND INTERACTION DESIGN

3.1 Simulated Annotation Process

Between two abutting virtual Tokyo, a simulation installation is depicted in Figure 3. This installation creatively visualizes the image data annotation process through physical simulation, consisting of a central label cube and three giant eyeballs rotating around it which launch images with physical simulation properties. These images rotate, fly, and block each other, eventually colliding with static labels in an attempt to match. Most randomly launched images bounce off and fall away due to mismatches, while only a few successfully match and attach to the corresponding labels.

The continuously emitted images echo Saussure’s concept of the signifier or sound-image[de Saussure 1916], representing the raw data from the real world awaiting annotation. The static labels, defined explicitly by text, correspond to the signified, the concept. Thus, this entire physical simulation not only reflects the AI data annotation process but also mirrors the relationship between the signifier and signified. Every match or mismatch established through image-label collisions is immediately recorded in a key-value pair, where the signifier serves as the key and the signified as the value (stored matching relationships), and this record is updated in real-time with each collision.

Participants can use the left-hand VR controller to grab any emitted but unmatched image in distance and place it anywhere. Notably, if a participant places an image on the correct matching label, the label turns blue and saves the correct match in stored matching relationships. However, if the participant casually discards the image, the label turns yellow but still records the participant-defined incorrect match relation. This interaction is designed to give participants a sense of how manipulating data annotation works. It highlights that finding the correct label manually in a complex, dynamic simulation with 600 categories is no easy task, and careless annotations can easily introduce bias into the data.

3.2 Distortion of Urban Data

The interaction simulation process defined in Section 3.1 essentially creates and maintains the stored matching relationships, a dynamic mapping of signifier to signified within a space of 600 entity categories. Unlike training data for real-world AI algorithms that strive to accurately identify and classify images, the participant-driven data annotation process introduces chaotic matching relationships. For instance, a chair might be labeled as a shoe, or a building paired with a dress.

In the VR environment, participant can freely navigate and fly over Tokyo with a small floating eyeball, who can also load street view images of any location by throwing the eyeball, as shown in Figure 4. When the eyeball lands in Tokyo, it retrieves the latitude and longitude of the collision point and loads the nearest street view panorama. The panorama is then split into six static images



Figure 4: Interactive pipeline for AI distorted street-view.

for six faces, which are arranged into a high-resolution immersive street cuboid surrounding the participant.

Similar to Rosenfeld's transplanted object experiment[Rosenfeld et al. 2018], we introduce AI-driven distortion by overlaying images on specific areas of the front, back, left, and right street view images. Specifically, for each street view image, we use an object recognition AI algorithm to identify all objects and their bounding boxes, assigning them one of 600 categories. The recognized category, or "signified," is then used to retrieve its corresponding "signifier" from the stored matching relationships, and the image from the built-in dataset associated with the "signifier" is overlaid onto the object's bounding box. Translating image data into objects is exactly how computer vision AI models such as YOLO perceive the world. Reversing these translated objects back into distorted images that humans can intuitively understand through

user-customized mapping relationships is the core of this section's interactive visualization.

Imagine skyscrapers in Tokyo being replaced with black-and-white low-rise buildings or a bar filled with various hippos due to a user-defined correspondence between hippo images and chair labels. In the VR panoramic space, these replacements amplify the visual impact of AI annotations, clearly illustrating how data manipulation affects AI perception.

3.3 Implement Details

The project is primarily built using the UE5 engine and python, leveraging Photo-realistic 3D Tiles and Street-View data for urban visualization. Additionally, part of the Open Images V7 dataset with bounding boxes is used for simulating the image training process. YOLOv8 is employed as the object detector, and GPT-4o is used to generate description for the distorted street-view images.

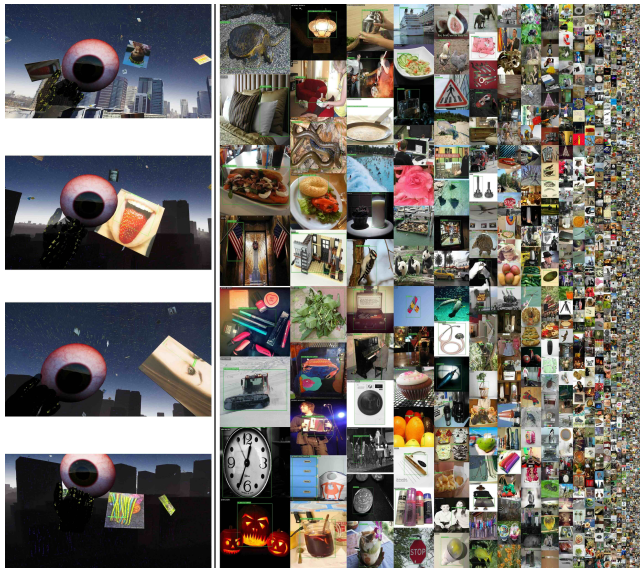


Figure 5: The image data used in the simulation consists of annotated images from the Open Images V7 dataset [Kuznetsova et al. 2020]. We selected 12 images from each of the 600 categories and added annotations which are displayed on the right side.

4 DISCUSSION

The understanding of inputs by AI systems is derived from their structure, training methods, and data, almost all of which are defined by humans. Particularly, the vast amounts of labeled data used for training reflect collective human intelligence, with each annotation carrying the imprint of human-centered cognition. This labeling process and its reverse (where AI’s outputs influence human cognition) constantly occur worldwide. This is symbolized in the work by the simulation of images being transmitted and colliding. Regardless of whether participants actively manipulate images to influence the training process, the giant eyeball representing human cognition continues to function as pre-defined, embedding human understanding into AI cognition through data annotation.

The second perspective examines how AI expresses its understanding of the world and the symbolic meaning behind these expressions. We assume that such expressions are presented in a human-understandable manner such as predicted labels, bounding boxes, coherent text, and so on. How AI generate output affects how AI encodes inputs [Huh et al. 2024]. This implies that when we train AI to produce expected human-like results, we also, in a sense, force AI to understand the world in a human-like way. In this work, we adopt an innovative perspective to express AI’s cognition of the world by having AI-driven distortion images of Tokyo streets.

This work also seeks to echo Michel Foucault’s metaphor of the Panopticon as a symbol of modern disciplinary society [Foucault 1995]. In the Panopticon, every prisoner is constantly visible, but they cannot see their observers, leading to a state of perpetual discipline and obedience. Similarly, AI systems we are familiar with rely on a broader set of political and social structures. These AI systems, continuously trained and optimized yet unable to perceive the full

scope of the world, mirror the Panopticon’s prisoners. Humans, positioned at the center of this metaphorical prison, project their understanding of the world, overseeing and shaping AI cognition. As AI becomes increasingly integrated into all aspects of human life, it also begins to influence human perception of the world, much like an inverted reflection of Tokyo. In this sense, AI serves as an imprint of power, simultaneously reflecting and shaping social relations and our understanding of the world. It acts as a mirror, faithfully representing humanity itself.

5 CONCLUSION

We developed “Abutting Tokyo” to explore AI cognition and the interactive visualization of urban data in VR environment. By incorporating a simulated image training process and AI-distorted landscapes of Tokyo, participants can experience AI’s perception and reconsider the role of humans in AI cognition. This work facilitates a discussion on AI’s response paradigms when faced with continuous real-world spatial data. We highlights the impact of human-centered data annotation on AI cognition, emphasizing that AI perception mirrors collective human cognition, reflecting our understanding of ourselves and the world.

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